AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_61906A98EC
Taxa	Allosaurus
Clade	Theropod
PBDB Records	147
Mass (regression)	2159.9 kg (Campione & Evans 2012)
Mass (volumetric)	1645.7 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	1.936 m³
Validity Score	1.0000 / 1.0000
TCI	0.6950 (partial)
Axiom I Entropy	0.9684
Axiom II λ■	0.1763 (symmetry 100%)
Axiom III Uniformity	0.6377
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	56.0% Measured / 44.0% Inferred
Elements	25
CoG (m)	[0.374, 0.837, 0.0]
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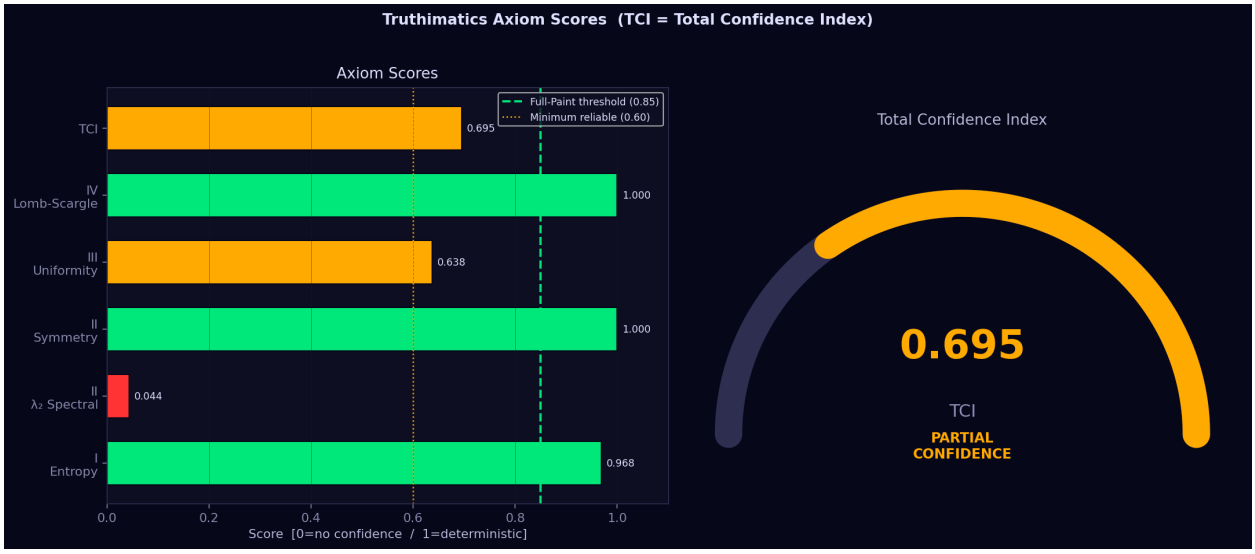
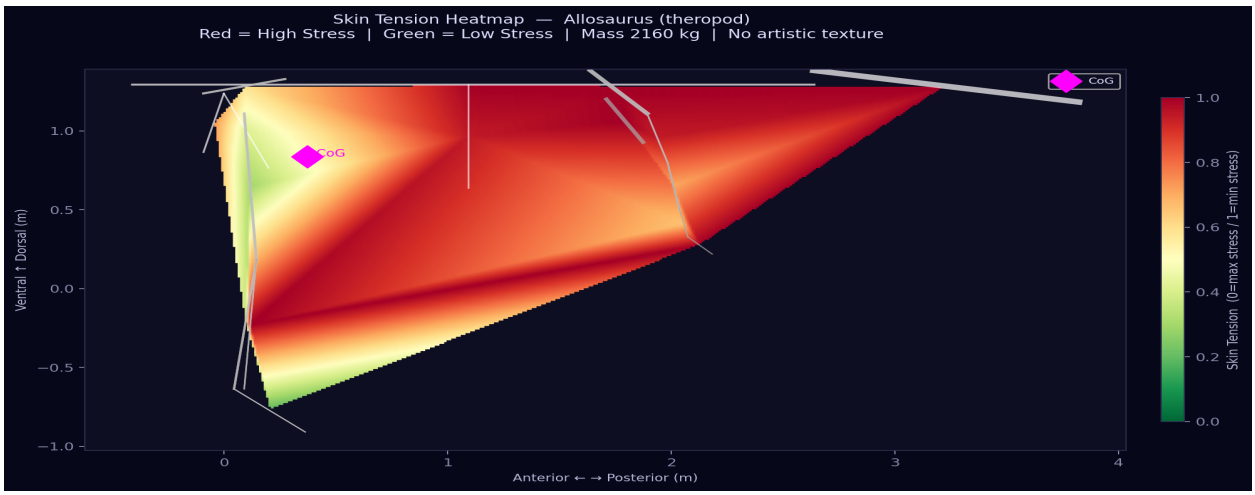
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



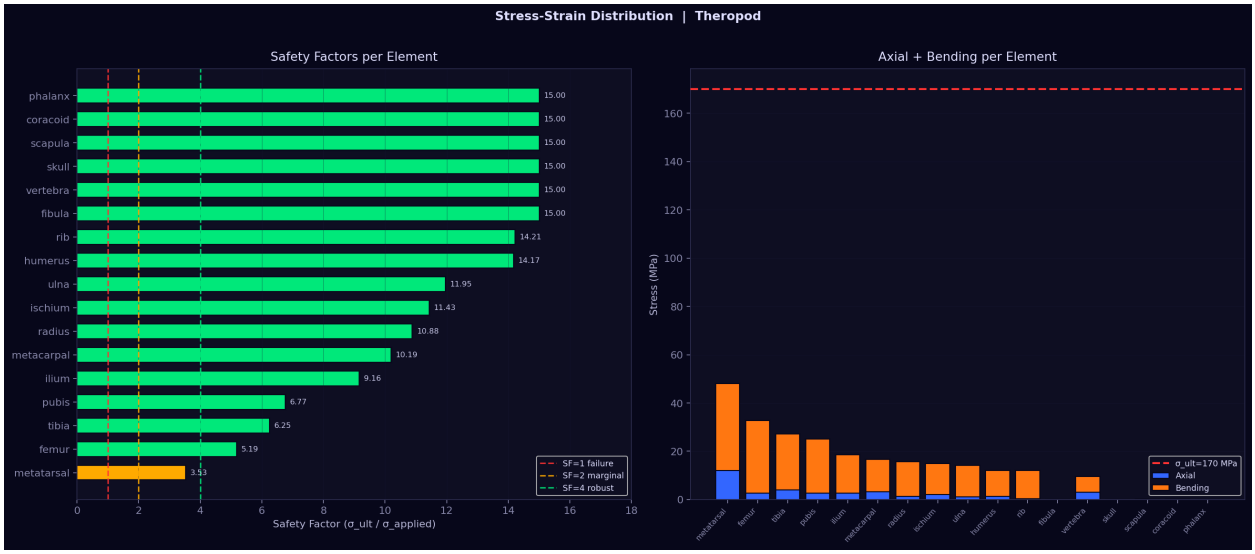
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

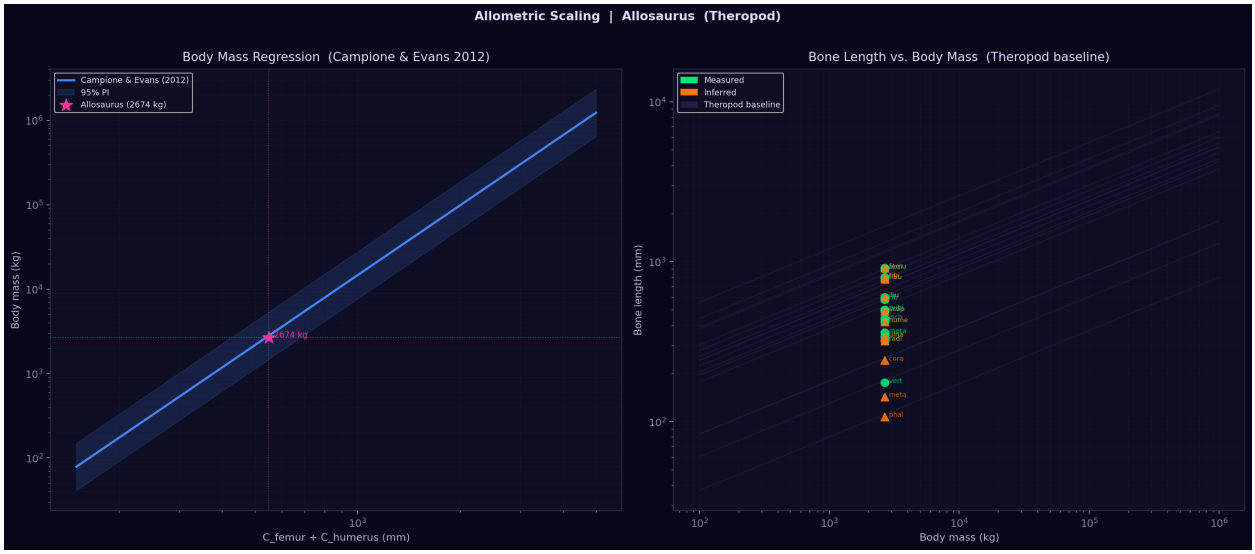


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.061	0.113	0.174	15.000	OK
femur	2.706	30.052	32.759	5.189	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	1.240	10.756	11.996	14.171	OK
ilium	2.706	15.852	18.558	9.160	OK
ischium	2.038	12.836	14.874	11.429	OK
metacarpal	3.285	13.391	16.676	10.194	OK
metatarsal	12.111	36.013	48.124	3.533	WARNING
phalanx	0.000	0.000	0.000	15.000	OK
pubis	2.713	22.408	25.121	6.767	OK
radius	1.270	14.355	15.625	10.880	OK
rib	0.348	11.614	11.962	14.212	OK
scapula	0.051	0.176	0.227	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	4.043	23.154	27.197	6.251	OK
ulna	1.146	13.077	14.223	11.952	OK
vertebra	2.970	6.645	9.615	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	2.07 m	Femur + tibia + metatarsal
Estimated Speed	16.5 km/h (4.57 m/s)	Gait: Trot | Fr=1.029
Stride Length	4.788 m	Alexander (1976) / Thulborn (1990)
Bite Force	9354.8 N (953.6 kgf)	Calibrated theropod regression (Bates & Falkingham 2012)
Skull Length	900.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.3798	Circumference / length (dimensionless)
Limb Strength Index	4.7e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	7.78 m	Allometric from femur length
Est. Neck Length	1.09 m	Allometric from femur length
Est. Tail Length	2.96 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	241.0	173.9	2917.39	3641.8	Constraint	0.40	15.00
femur	910.0	110.0	4410.49	5505.6	Measured	0.88	5.19
femur_right	910.0	110.0	4410.49	5505.6	Symmetry	0.84	5.19
fibula	780.0	38.0	318.46	334.0	Measured	0.88	15.00
fibula_right	780.0	38.0	318.46	334.0	Symmetry	0.84	15.00
humerus	420.0	65.0	710.78	887.3	Measured	0.88	14.17
humerus_right	420.0	65.0	710.78	887.3	Symmetry	0.84	14.17
ilium	600.0	110.0	2908.02	3630.1	Measured	0.88	9.16
ilium_right	600.0	110.0	2908.02	3630.1	Symmetry	0.84	9.16
ischium	440.0	75.0	991.37	1237.5	Measured	0.88	11.43
metacarpal	142.8	38.7	31.93	26.3	Constraint	0.65	10.19
metatarsal	360.0	52.0	389.91	486.7	Measured	0.88	3.53
phalanx	107.1	36.2	16.95	13.1	Constraint	0.40	15.00
pubis	500.0	65.0	846.17	1056.3	Measured	0.88	6.77
radius	320.0	38.0	185.09	231.0	Measured	0.88	10.88
radius_right	320.0	38.0	185.09	231.0	Symmetry	0.84	10.88
rib	580.0	28.0	182.14	227.4	Measured	0.88	14.21
scapula	490.0	190.0	7085.38	8844.7	Measured	0.88	15.00
scapula_right	490.0	190.0	7085.38	8844.7	Symmetry	0.84	15.00
skull	900.0	420.0	63591.80	79381.6	Measured	0.88	15.00
tibia	800.0	90.0	2595.58	3240.1	Measured	0.88	6.25
tibia_right	800.0	90.0	2595.58	3240.1	Symmetry	0.84	6.25
ulna	340.0	40.0	217.90	272.0	Measured	0.88	11.95
ulna_right	340.0	40.0	217.90	272.0	Symmetry	0.84	11.95
vertebra	175.0	105.0	772.82	964.7	Measured	0.88	15.00